

Charles Villard Ph.D.

villard.it

charles@villard.it | +33 650818590

LIENS

Github:// **Silvanosky**
LinkedIn:// **charlesvillard**
ResearchGate:// **Charles-Villard**
ORCID:// **0000-0003-3636-3850**

EDUCATION

UNIVERSITY GUSTAVE EIFFEL
MATHEMATICS AND INFORMATION
AND COMMUNICATION SCIENCES
AND TECHNOLOGIES.

PhD in Underwater Exploration
Robotics and Photogrammetry.
2020 - 2024 | Paris

EPITA
COMPUTER SCIENCE ENGINEER
Specialised in real-time embedded
systems. 2015 - 2020 | Paris

STELLENBOSCH UNIVERSITY
ABROAD SEMESTER IN COMPUTER
SCIENCE AND ELECTRONICS. 2017
| South-Africa

SKILLS

AI & Optimization

Generative AI, SOTA Research,
Anomaly Detection, Quantization,
Tensor Decomposition, Model Compression,
PyTorch, CUDA, TensorRT, Data
Augmentation

Embedded & Edge

Edge AI, Real-time Systems,
Eclipse Aidge, ARM, ESP32,
FreeRTOS, Yocto, Buildroot,
AOSP, C, C++, Assembly

Computer Vision

Photogrammetry, SLAM,
3D Reconstruction, OpenCV,
MicMac, Point Clouds

Development

Docker, Git, CMake, Python,
VHDL, Java, SQL, Shell

SPOKEN LANGUAGE

French (Native)
English (Fluent)

HOBBIES

Air: FPV Freestyle drone pilot, DIY RC
Planes, 3D Print

Water: Scuba Diving – CMAS 1* Instructor /
Dive Guide (60m)

Ground: Climbing, Speleology

SUMMARY

Ph.D. Researcher specializing in the intersection of **Embedded Systems** and **Deep Learning**. Currently leading research at CEA List on **Generative AI** for industrial anomaly detection and **Hardware-Aware Model Compression** for edge deployment. Expert in bridging complex CV algorithms with resource-constrained hardware.

EXPERIENCES

CEA List | POST-DOC IN DEEP LEARNING FOR EMBEDDED SYSTEMS

July 2024 - July 2027 | Palaiseau, FR

- **Generative AI for Anomaly Detection:** Engineering SOTA generative models to synthesize high-fidelity data, mitigating class imbalance and enhancing industrial detection performance (Research in progress).
- **Edge AI & Model Compression:** Architecting hardware-aware optimization (Tensor Decomposition, Quantization) for complex NN deployment on the Eclipse Aidge platform within the DeepGreen (France 2030) project.

EPITA/LRE - IGN/LaSTIG/ACTE | PH.D IN ROBOTICS AND PHOTOGRAMMETRY

Nov 2020 - Apr 2024 | Paris, FR

- Engineered a **hierarchical Computer Vision pipeline** for 3D reconstruction, optimizing spatial data processing for low-power underwater compute modules.
- Designed and deployed an autonomous underwater acquisition robot and its payload: a low-cost synchronized vision system.

Safran Electronics & Defense | END OF STUDY INTERSHIP

Fev - Sept 2020 | Massy, FR

- Developed a hardware/software localization stack for **autonomous vehicles**, integrating IMU/GNSS data in a real-time C++ environment.

EPITA research laboratory | RESEARCH UNDERGRAD STUDENT

Jan 2018 - Sept 2020 | Paris, FR

- Development of multi-domain robots (ground, air, sea). Software/hardware architecture and integration of vision-based sensors.

LIRMM | SOFTWARE ENGINEER INTERSHIP

Sept - Dec 2018 | Montpellier, FR

- Building a real-time software architecture for a modular underwater robot designed to map complex underwater cave systems.

PUBLICATIONS

- [1] **Villard, Charles**, *Acquisition using lightweight platforms and data processing for underwater photogrammetry*. Theses, Université Gustave Eiffel, 2024.
- [2] **Villard, Charles**, E. Rupnik, and M. Pierrot-Deseilligny, "Estimation initiale de la pose avec un Échantillonnage aléatoire mais pondéré de paires de triplets redondants d'images," in *ORASIS 2023, Laboratoire LIS, UMR 7020*, 2023.
- [3] **Villard, Charles** and et al., "Automatically guided selection of a set of underwater calibration images," *Journal of Marine Science and Engineering*, vol. 9, no. 5, p. 740, 2022.
- [4] **Villard, Charles**, L. Avanthey, L. Beaudoin, and et al., "Synchronization of PiCam cameras for three-dimensional study of dynamic multi-domain natural scenes," in *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. V-1-2020, pp. 277–284, 2020.
- [5] **Villard, Charles**, L. Beaudoin, L. Avanthey, and et al., "Porting Ardupilot to ESP32: Towards a universal open-source architecture for agile and easily replicable multi-domain mapping robots," *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLIII-B2-2020, pp. 933–939, 2020.

TEACHING & PROJECTS

Lecturer & Course Head (EPITA): Currently leading the **Embedded Linux Distribution (Yocto)** course. Previously assisting for **C++, Algorithmics, Computer Vision, and Robotics** (2018–Present).

Portfolio: villard.it/cv/

Highlights: Ardupilot ESP32 porting, Photogrammetry pipelines, personal projects.